Outskill

Multimodal Fusion & Embeddings

Recap of last week

What have I learnt?

1. Introduction
2. Working
3. Advantages
4. Evolution
5. Applications
6. Challenges
7. Multimodal Data Representation (Code)
8. Individual Modality Processing (Code)
9. Evaluation (Code)
10. Real world multimodal datasets
11. HuggingFace Datasets Package
12. HuggingFace Transformers Package
13. PyTorch
14. PyTorch Dataset and DataLoaders
15. Multimodal Data Alignment

What will I learn?

1. Embedding Modalities
2. Dimensionality Reduction Techniques
3. Attention and Cross Attention
4. Multimodal Fusion Strategies
5. Modality Dropout

Code

Image emb: https://colab.research.google.com/drive/1Pet6oTv4m9QMl0Edz5-YDIVuCQiLsd0s?usp=sharing

Audio emb: https://colab.research.google.com/drive/1-CNQToo4-a2DiDdKkQS5Md8cS8_6Ospn?usp=sharing

Text emb: https://colab.research.google.com/drive/10VgClUleNE6dAEd1MaaHYd7HHQjNBw0m?usp=sharing

Late fusion: https://colab.research.google.com/drive/18qFm1wdS0e7Z_WKXWY5fL3F0hRFLV245?usp=sharing

Breakout rooms: https://colab.research.google.com/drive/1pyo7SQFUY1Kz6fPLCitUZO5rYl4eDKmG?usp=sharing

Embedding Modalities

1. An ML model that performs really well in english.
2. We’ve served the model to our users in production
3. The users demand for the model to understand “hindi” as well.
4. A research engineer post-trains this model on hindi data (RL based, SFT etc.), and he manages to get good perf on hindi datasets
5. So we serve it but we notice that our serving costs are 5x. why?

What are Embeddings? And encodings?

Embeddings are numeric vectors that encode the semantic meaning of a modality.

Semantic analysis involves interpreting the meaning of data, such as text or images, to extract relevant information and insights. It considers the context in which data is presented, not just individual words or features.

Semantic analysis helps machines understand the intent behind user queries or commands, rather than just matching keywords.

[Embedding Projector]

Embeddings are learnt! Encodings aren't and hence they're not semantic.

Encodings.

Just a way to encode stuff!

Let’s understand via one hot encoding (we’ve already done this btw) –

Cat - [1, 0, 0]

Car - [0, 1, 0]

Motor - [0, 0, 1]

Text

Tokens → Transformer → Vector

Image

Pixels → CNN/ViT → Vector

Audio

Waveform → Spectrogram → CNN/RNN → Vector

Any type of data -> emb model (that can work with that data type) -> embedding

Dimensionality Reduction

- **Definition**: A technique to reduce the number of features in a dataset while retaining its meaningful information.
- **Why it's needed:** Simplifies models, improves performance, and reduces computational cost.
- **Key Applications:** Visualizing high-dimensional data, noise reduction, and faster machine learning models.
- **Types:** Linear (e.g., PCA) and non-linear (e.g., t-SNE, UMAP).
- **Challenges:** Choosing the right method and balancing information loss.

5	7	2	-1	6	7	4	6	7	2	7	2	7	8	3	5	6	0	1

Goal of the emb model is to generate embeddings
What is a good emb model? That clusters similar data points together.
MNIST DATA: A good img emb model should generate embeddings such that imgs of the same number have embeddings that are closer to one another

Principal Component Analysis

- **Definition:** A linear dimensionality reduction technique that transforms data into uncorrelated variables (principal components).
- **How it works:** Maximizes variance captured in fewer dimensions.
- **Steps:** Centering data, calculating covariance matrix, eigen decomposition, and projection.
- **Use Cases:** Feature extraction, data compression, and exploratory data analysis.
- **Pros/Cons:** Efficient for linear data but less effective for complex, non-linear relationships.

[PCA Demo]

Rows represent data point and cols represent dims

				2.2						10k		
				2.01						-0.9		
				2.5						-20k		
				2.4						32		
				2.21						5k		
				2.34						- 0.00 3		
				2.3						1		

X1 … x13

Step 1: transform the orig var system into a diff system
Y1 … y13 - pca components

Step 2: y1 = x1 + 2x2 + 0.5x3 + … x13
Y2 … y13

The way y1..y13 are constructed is st y1 has the hight variance followed by y2 and so on
Var (y1) > var(y2)...

PCA

X1 x2 x3 x4 – original dims

New dims - y1 y2 y3 y4 - PCs

t-SNE (t-Distributed Stochastic Neighbor Embedding)

- **Definition:** A non-linear technique for visualizing high-dimensional data in 2D or 3D spaces.
- **How it works:** Preserves local structure by modeling pairwise similarities with probabilities.
- **Strengths:** Great for visualizing clusters in datasets.
- **Limitations:** Computationally intensive and not suitable for preserving global structure.
- **Applications:** Embedding spaces, gene expression data, and image clustering.

Heatmap

- **Definition:** A graphical representation of data where values are represented by colors.
- **Common Use:** Visualizing correlations, distributions, and relationships in data matrices.
- **Construction:** Rows and columns correspond to features or observations; colors indicate value intensities.
- **Applications:** Feature importance, gene expression data, and clustering patterns.
- **Advantages:** Intuitive visual summary, but less detailed for precise data analysis.

Attention

Attention in Transformers

- **Core Idea:** Attention allows the model to focus on the most relevant parts of the input data when understanding the meaning of a specific word or token in the sequence.
- **Analogy:** Imagine reading a sentence where some words are more important to understand the current word. Attention works like highlighting those important words while reading.
- **Self-Attention:** Each word in the input looks at all the other words to decide which ones are important to its context, enabling the model to understand relationships between words.
- **Dynamic Focus:** Unlike older methods, attention dynamically adjusts its focus based on the context of the input, making it better at capturing complex relationships.
- **Applications:** It is the backbone of tasks like language translation, question answering, and summarization, allowing the model to process entire sequences efficiently.

[Attention Demo]

Cross Attention

- **Core Idea:** Cross-attention allows the model to focus on relevant parts of a different input sequence when generating an output, such as aligning text with images or translating between languages.
- **Difference from Self-Attention:** Instead of focusing on the same input (as in self-attention), cross-attention looks at one sequence (e.g., a text description) while processing another (e.g., image features).
- **How It Works:** It learns to link and prioritize information from one sequence (source) to enhance understanding and processing of another sequence (target).
- **Dynamic Alignment:** Cross-attention dynamically identifies which parts of the source sequence are most relevant to each part of the target sequence, improving tasks like image captioning or multimodal learning.
- **Applications:** text-to-image generation, CLIP, etc

Multimodal Fusion Strategies

What is Fusion?

- **Multimodal fusion** refers to the process of **combining** information from **multiple modalities** in order to enhance the understanding or analysis of a certain phenomenon or problem.
- In essence, it involves **integrating data** from different sources, such as text, images, audio, video, and sensor readings to gain a more **comprehensive** and **accurate representation** of the underlying information.

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Early

1. Fusion Stage: Input Level
2. Mechanism: Concatenate Raw Features
3. Use Case: Sentiment

Intermediate

1. Fusion Stage: Latent Space
2. Mechanism: Concat/add/attention on embeddings
3. Use Case: Autonomous Vehicle

Late

1. Fusion Stage: Output Level
2. Mechanism: Averaging/Voting on outputs
3. Use Case: Healthcare

Hybrid

1. Fusion Stage: Multiple
2. Mechanism: Combination of the others
3. Use Case: Video-QA

Early Fusion

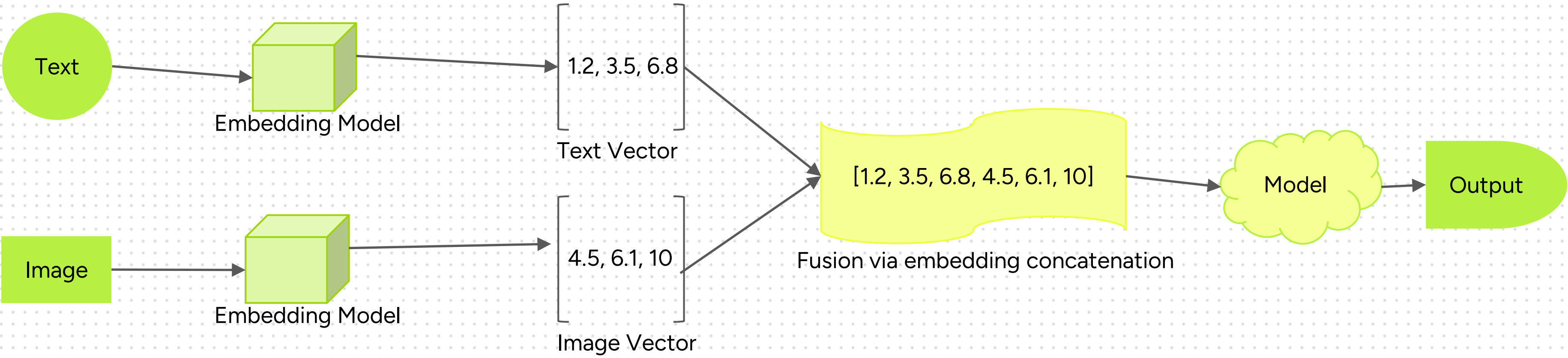
Input-level fusion: Combine raw or lightly processed data before model.

Example: Text of review + metadata (price, author) concatenated for sentiment analysis.

Pros: Simple, single model, unified pipeline.
Cons: High-dimensional input; modality-specific nuances may be lost.

Early Fusion Example

[Image + Text Classifier]



Intermediate Fusion

Encode each modality separately; fuse in hidden layer. Fusion via concatenation, addition, or attention.

Example: Autonomous driving — camera, LiDAR, radar embeddings fused before decision-making.

Pros: Rich intra-modal interaction.
Cons: Computation-heavy; requires separate encoders.

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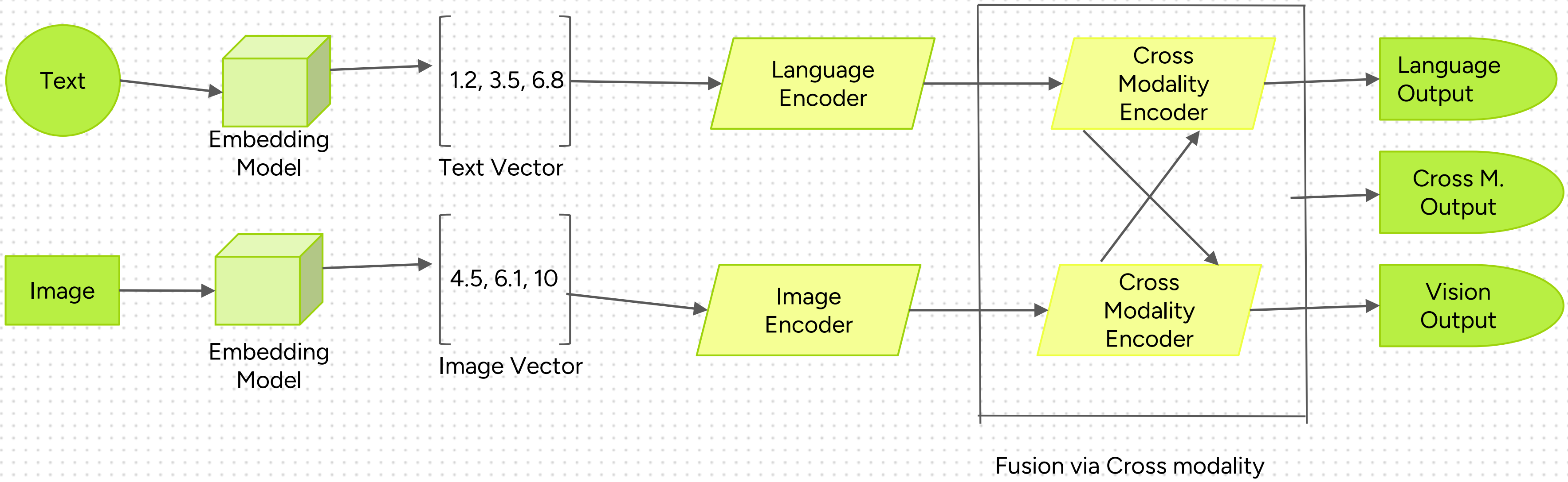
Intermediate Fusion Example

[LXMERT Model]



Late Fusion

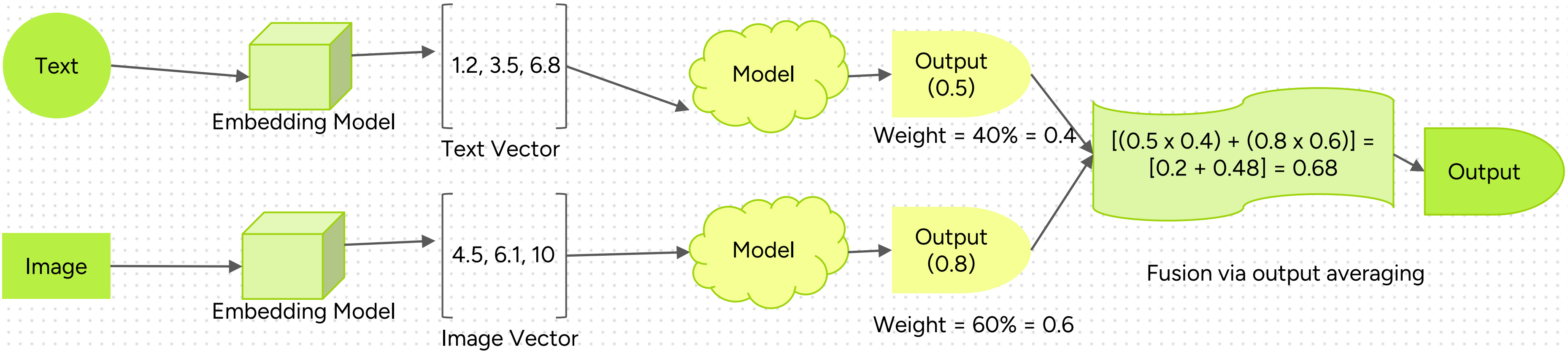
Train separate pipelines for each modality. Fuse via score/vote.

Example: Video genre classification with transcript and visual model; fuse predictions.

Pros: Modular, interpretable, robust to missing modalities.
Cons: Misses fine-grained interactions; may need additional logic to combine outputs properly.

Late Fusion Example

[Product Classifier]



Hybrid Fusion

Combines elements of two or more fusion strategies.

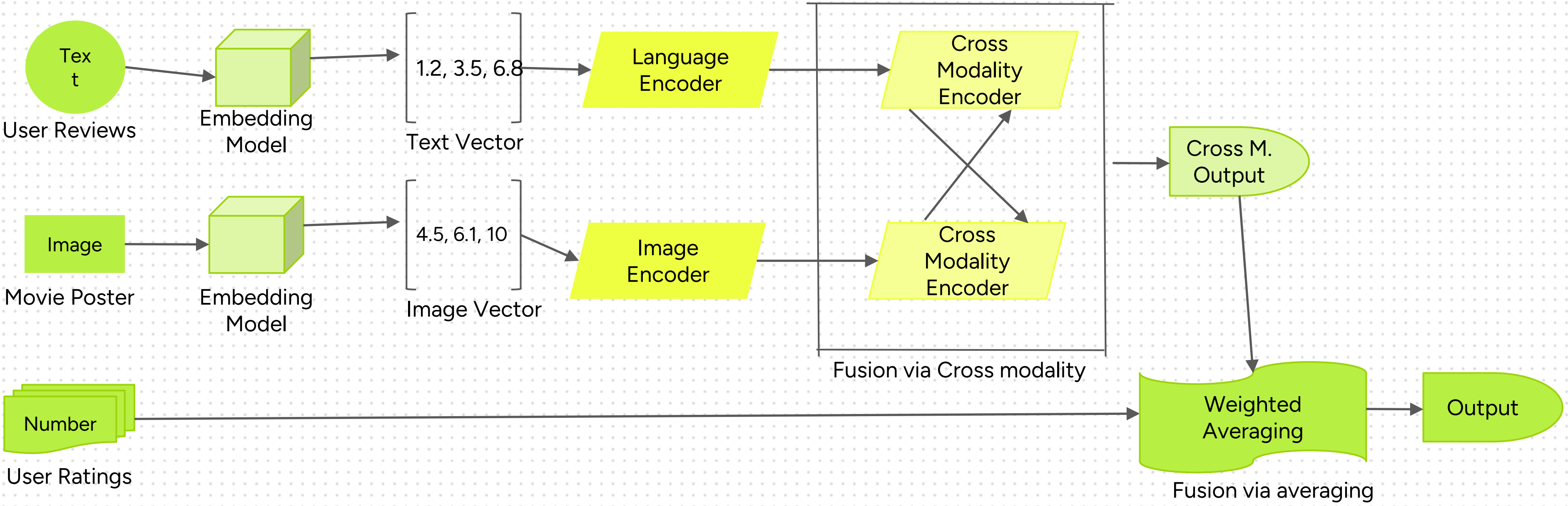
Example: Video QA:

- Text+video early embedded with cross-attention.
- Outputs later fused with metadata classifier.

Pros: Balances inter-modality modeling and modular flexibility.
Cons: Increased complexity in architecture and training.

Hybrid Fusion Example

[Movie Recommendation]



Modality Dropout

- Ideally, we would want to design a **system** which could **integrate all** the available **modalities**, and still **function** effectively if certain **modalities** are **unavailable**.
- One strategy to accomplish this is through **modality dropout**. Modality dropout (or modal dropout) involves randomly **dropping** or **obscuring** certain modalities during each training iteration, forcing the model to adapt to varying combinations of modalities.
- This enables the model to effectively utilize the available modalities, so that even if only one of the modalities are available, the model can still produce reasonable predictions.